

Q6(a)(i)

In one minute, the energy absorbed by the water is $E = Mc\Delta\theta = (20)(4200)(3.0)$
= 252 kJ

Q6(a)(ii)

$$\begin{aligned}\text{Solar power per square metre} &= \frac{E}{tA} = \frac{4E}{t\pi d^2} \\ &= \frac{4(252 \times 10^3)}{(60)(3.142)(4.0^2)} \\ &\approx \mathbf{334 \text{ W m}^{-2}}\end{aligned}$$

Q6(a)(iii)

Thermal energy is lost from the pool and so the measured temperature rise is too low. This leads to measured energy and power values which are too low.

Q6(b)(i)

$$\begin{aligned}\text{Power per square metre is } \frac{VI}{A} &= \frac{(7.0)(200 \times 10^{-3})}{(0.15)(0.15)} \\ &\approx \mathbf{62 \text{ W m}^{-2}}\end{aligned}$$

Q6(b)(ii)

$$\begin{aligned}\text{Efficiency} &= \frac{\text{power of cell}}{\text{actual power}} \times 100 \\ &= \frac{62}{334} \times 100 \approx \mathbf{19\%}\end{aligned}$$

Q6(c)(i)

The Sun emits energy in all directions in the form of spherical wavelets.

$$I = \frac{P}{4\pi r^2} \text{ or } P = I(4\pi r^2) = (1.4 \times 10^3)(4)(3.142)(1.5 \times 10^{11})^2$$

$$= \mathbf{3.96 \times 10^{26} \text{ W} \approx 4.0 \times 10^{26} \text{ W}}$$

Q6(c)(ii)

Intensity of light absorbed by panels

$$I = 0.40 \times 1.4 \times 10^3 \text{ W m}^{-2}$$

$$I = \frac{P}{S} \text{ or } S = \frac{200}{0.40 \times 1.4 \times 10^3} = 0.357$$

$$\approx \mathbf{0.36 \text{ m}^2}$$

Q7(a)